IMMERSION DEPOSITION OF CU ON STEEL

MME408

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Introduction:

Immersion or displacement coating is a process of dipping a cathode into a metal solution to form a thin metal layer on the substrate. It is an old technique of metal deposition, more specifically, ancient civilizations like the Phoenicians and Carthaginians used copper sheathing for boats, demonstrating early forms of protective coatings. Also in the early 9th century, Europeans used displacement plating to coat iron armor with copper, preparing it for gilding. Even though this technique is used to give a basic idea about deposition & coating, it is also used for small-scale (laboratory) production.

When depositing Cu on Fe, it is deposited rapidly because it has a large E^0_{Fe}(oxidation), which will cause Fe to dissolve (act as an anode) and Cu to deposit (act as a cathode)

From the standard EMF series:

$Cu = Cu^{2+} + 2e^-$; $(E\ )_{ox} = -\ 0.337\ V$——–—(1)

$Fe = Fe^{2+} + 2e^-$; $(E\ )_{ox} = +\ 0.440\ V$——–—(2)

$Cu^{2+} + 2e^- = Cu$; $(E\ )_{red} = +\ 0.337\ V$——–—(3)

$\Delta G = -\ E * n * F$; $\Delta G = -$ ve; Reaction will occur (spontaneous).

Reaction (1) is non-spontaneous, but reactions (2) & (3) are spontaneous.

$Fe = Fe^{2+} + 2e^-$;$(E)_{ox} = +\ 0.440\ V$——–—(2)

$Cu + 2e^- = Cu$;$(E)_{red} = +\ 0.337\ V$——–—(3)

$Fe + Cu^{2+} = Cu + Fe^{2+}$; $(E)_{total} = +\ 0.777\ V$——–—(4)

Reaction (4) is spontaneous.

As soon as steel is deposited into the $CuSO_4$ bath, Cu will be deposited onto the steel. This immersion process will be less effective due to the depositing non-adherent or spongy deposit will form.

When we give a quality coating, we can use a complex Cu bath (Cu-cyanide bath), where we should use external energy to make the process spontaneous, and also the deposition time will be lower. Additionally, it has excellent adherent properties.

The objectives of this lab :

1. To be introduced to the basics of Immersion deposition of Cu on steel
2. To be familiar with the old deposition techniques.
3. How time and composition of the Cu bath affect the final product of deposition.
4. To know about the surface morphology of the metal on the substrate
5. To acknowledge the weight of metal on the substrate at a fixed time.

Methodology:

1. First, we collected the steel sample, emery paper, plastic substrate holder, clamp, and vernier scale from the lab assistant.
2. Then we rub the sample on the emery paper and blunt its edges.
3. We washed the sample with detergents and gently handled it afterwards.
4. Then we took the weight of our sample and noted it.
5. We took the dimensions (height, weight, thickness) of our sample and noted that.
6. We gently marked 3.6cm (length of deposition) from the bottom of the sample (opposite of the hole).
7. We then immersed the sample in a pickling medium (20% HCl)(Fig.3)
8. Then we rinsed it with tap water and dried.
9. After that, we measured its weight again.
10. Then we carefully placed it inside 200mL $CuSO_4$ solution(Fig4) and started our clock for 10 mins.
11. After 10 mins, we dried the sample over a hot plate and weighed and measured its thickness for further procedures.

Figure 2: Plastic Substrate Holder

Figure 1: Clamp

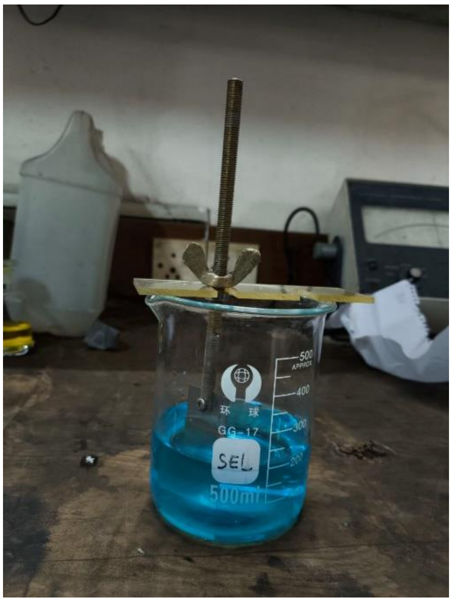

Figure 4: Deposition of Cu on Steel

Figure 3: Pickling of Steel Substrate

Calculation & Results:

Length of deposition = 3.6 cm

Width = 2.9cm (with and without deposition)

Thickness (before deposition) =0.055cm

Thickness (after deposition) = 0.1cm

Thickness of deposited Cu = (0.1-0.055) cm = 0.045cm

Weight before pickling = 8.54gm

Weight after pickling = 8.34gm

Weight after deposits = 8.4gm

Weight loss during pickling = (8.54-8.34) gm= 0.2gm

Weight of Cu deposits = (8.4-8.34) gm= 0.06gm

Cu deposition per unit area = $\frac{0.06gm}{(3.6 \times 2.9 \times 2)cm^2}$ **= 0.00287 gm/cm^2**

Density of Cu = 8.96 gm/cc

Volume = (0.045 * 2.9 * 3.6) cc = 0.4698 cc

The deposited weight should be = (8.96 * 0.4698) = 4.21gm

It is different from the weight we calculated earlier, which will be discussed in the discussion section.

Discussion:

Why was emery paper used?

The steel sample might contain an oxide or other impurity layer on the outside of our substrate, which could damage our final coating. Also, these impurity layers could make the adherence property of Cu less effective. So, this outer coating must be removed before immersion. In addition, the edges are blunted to avoid accumulation of extra metal on the edges or uneven deposition.

What is the role of pickling in this case?

After using emery paper, there could still be some impurities. So, for proper cleaning and to expose the newly clear steel surface, pickling was done with 20% HCl.

Why did we use a length of 3.6 cm instead of taking the full length of 5.635 cm?

Because the surface where the deposit will form should be fully immersed in liquid. To examine the full surface effect, we would need more solution, as the substrate cannot touch the bottom of the beaker. In addition, we should cancel out the hole area while calculating deposition per area. In a lab experiment, it would be a waste of solution, so considering all these factors, our lab instructor instructed us to take a length slightly above half of the total length.

Surface morphology:

Our surface was clean and smooth. No holes were observed with the naked eye. After drying, the adhesion property was also good.

From our lab instructor, he told us that if we had increased the deposition time (at least 20 minutes), we could have gotten better surface quality.(Fig. 5) Moreover, no uneven deposition was observed at the edges.

Also, from the thickness difference, the weight difference, and the outer color, it is confirmed that Cu was deposited on steel.

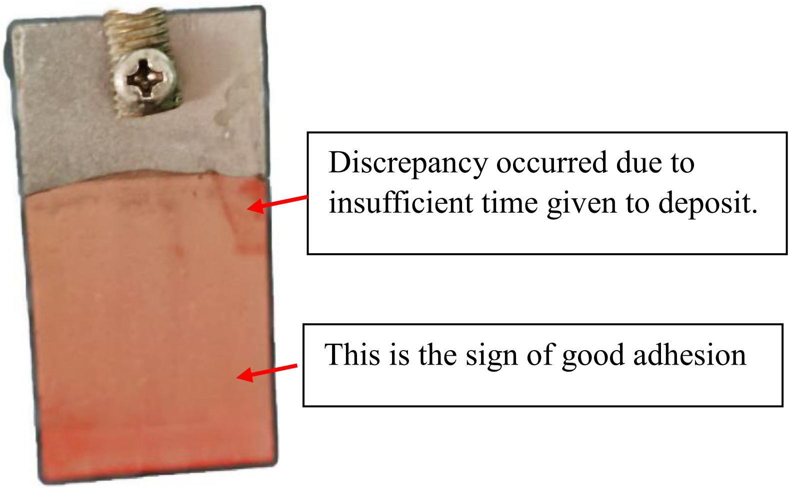

Figure 5: Deposition of Cu on Steel Substrate

Limitations of Simple Cu bath:

If a complex Cu bath (cyanide or sulphate bath) was used, the deposition would be more controlled. Additionally, within this short bath time (10 min), the deposition morphology would be much better. We would just need an external current supply to make the process spontaneous.

Why was an external current supply not needed?

Here, Fe acts as the oxidizing agent and Cu^{2+} in solution acts as the reducing agent. It is a spontaneous process. So it conducts this reaction (unlike the complex bath), and no external current was needed.

Why did the difference between theoretical and practical value occur?

The deposited weight was 0.06g. But Cu can be deposited up to 4.21gm (from our calculation). Because proper time was not given to deposit this amount of Cu. Time could be calculated using the formula W= ZIt ; where internal charge transfer should be known.

Curve (Using deposited weight of different groups)

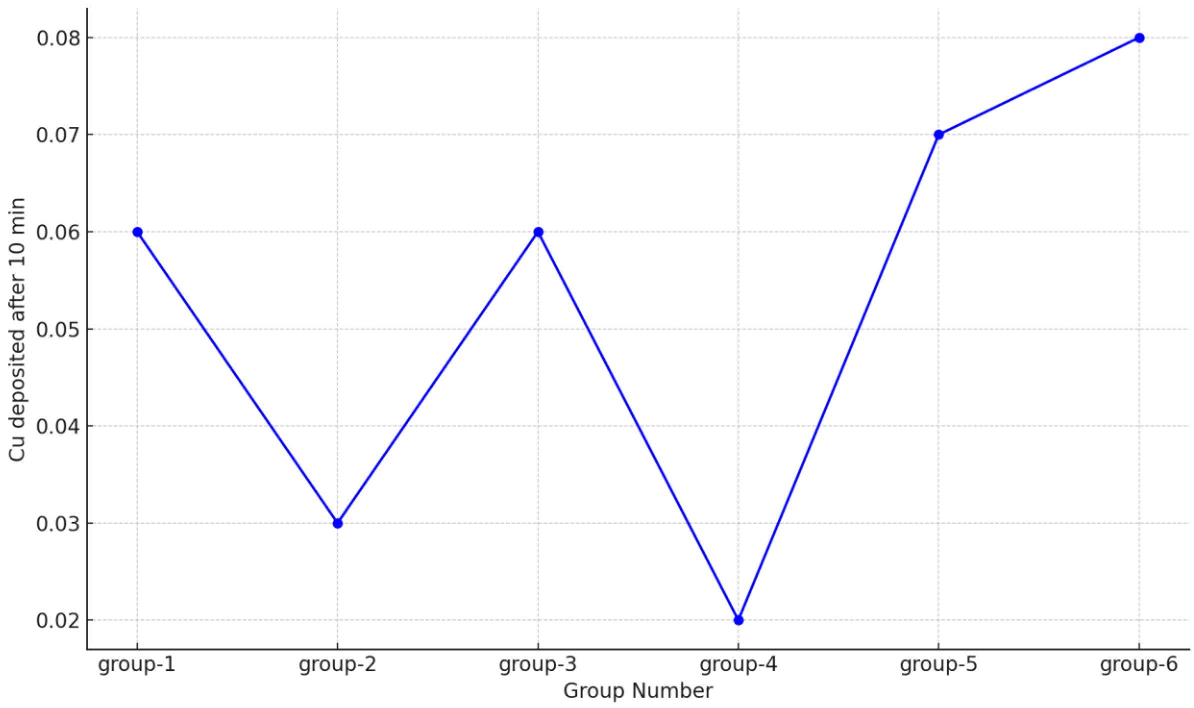

Figure 6: Data curve of different groups